Digital Electronics (ECT-033) - Unit 2

Unit 2 - Combinational Circuits - notes in easy English, matched to the UTU syllabus.

Study Materials by Anas - study-materials-by-anas.vercel.app

What this unit covers: combinational circuits - outputs depend only on the present inputs: adders, subtractors, multiplexers, demultiplexers, encoders, decoders, comparators.

Key ideas: Half adder: Sum = A⊕B, Carry = AB. **Full adder**: Sum = A⊕B⊕Cin, Carry = AB + BCin + ACin (two half adders + OR). A **MUX** (2^n:1) picks one of 2^n inputs using n select lines - any Boolean function of n variables fits in a 2^{n-1}:1 MUX with the remaining variable on the data lines (a favourite implementation question). A **demux** does the reverse; a **decoder** (n-to-2^n) activates one output per input combination - a decoder + OR gates implements any SOP function. An **encoder** compresses 2^n lines to n bits; a **priority encoder** resolves simultaneous inputs. A **magnitude comparator** outputs A>B, A=B, A<B.

Exam pointers: "implement f(a,b,c) using an 8:1 MUX" and "using a 4:1 MUX" are repeated numericals - practise the truth-table-to-data-lines mapping. Half/full adder truth tables with K-map derivations, and decoder-based function implementation, are standard theory.

Extra note - Barrel Shifter & ALU - made simple:

Barrel shifter: a circuit that shifts data bits left or right by **any number of positions in a single clock cycle**.

> *Like musical chairs - everyone gets up at once and sits 2 seats ahead. The whole shift happens in one round. A normal shift register moves only 1 seat per tick; a barrel shifter moves 2, 3, 4... whatever you ask, in one tick.*

How it is built: from rows of MUXes (multiplexers). Each output bit has a MUX that chooses which input bit lands there. The shift amount (how far to shift) drives the select lines of the MUXes.

ALU (Arithmetic Logic Unit): the part of a processor that does both **calculation** (add, subtract) and **logic** (AND, OR, XOR).

- Select lines decide the operation: 00 = ADD, 01 = SUB, 10 = AND, 11 = OR (example).
- Inside: full adders + logic gates + a MUX that picks the final answer.

Exam tip: "ALU = adder + logic gates + MUX" - write these 3 things with a diagram and the marks are yours.